

Eclipse 2025.4 — Dual Porosity / Dual Permeability

1-to-1 mathematical model, every equation traced back to the manual

Library: my-axon/libraries/eclipse-dual-porosity

Built 2026-05-22 from Eclipse 2025.4 Technical Description and Reference Manual

1 Notation and scope

Subscripts. m = matrix cell. f = fracture cell. i, j = grid indices. o, g, w = oil, gas, water phases.

Symbols.

σ shape factor, units $1/L^2$ (SIGMA, SIGMAV, LTOSIGMA)

K permeability used in the matrix-fracture coupling (matrix K_x by default, override via PERMMF)

V matrix cell *bulk* volume (no porosity factor)

CDARCY Darcy-unit conversion constant (≈ 0.00112712 in FIELD, ≈ 0.0086400 in METRIC)

Φ_p phase potential = $P + P_{c,p} - \rho_p g D$ for phase p

λ_p mobility = $k_{r,p}/\mu_p$

L_x, L_y, L_z physical matrix-block dimensions (LX, LY, LZ)

ΔZ_{mat} typical matrix-block height (DZMTRX, DZMTRXV)

ρ_p phase density at reservoir conditions

g acceleration of gravity

Z-doubling rule. The number of layers NDIVIZ (DIMENS item 3) must be even. Layers $1 \dots \text{NDIVIZ}/2$ are matrix; layers $\text{NDIVIZ}/2 + 1 \dots \text{NDIVIZ}$ are fracture. For multi-porosity with N matrix porosities (Eclipse 300), the rule generalises to NDIVIZ a multiple of $N + 1$.

2 Matrix–fracture transfer function (the core equation)

For every co-located (m, f) pair Eclipse builds an automatic non-neighbour connection (NNC). The matrix–fracture transmissibility is

$$T_{mf} = \text{CDARCY } \sigma K V \quad (\text{TD 2.54})$$

The phase flow rate between matrix and fracture is the upstream-weighted Darcy flux

$$q_{p,mf} = T_{mf} \lambda_p^\uparrow (\Phi_{p,f} - \Phi_{p,m}) \quad (1)$$

where $\lambda_p^\uparrow$ is the mobility evaluated in the upstream cell (determined by the sign of the potential difference, including any gravity head term, see §3).

Kazemi (1976) shape factor

For a matrix block with characteristic dimensions L_x, L_y, L_z :

$$\sigma = 4 \left(\frac{1}{L_x^2} + \frac{1}{L_y^2} + \frac{1}{L_z^2} \right) \quad (\text{TD 2.55 ; RM 3.172})$$

L_x, L_y, L_z are physical matrix-block dimensions, not simulation grid sizes. Setting $\sigma = 0.12 \text{ ft}^{-2}$ in *FIELD* units corresponds to a 10 ft cubic block.

LTOSIGMA form (anisotropic, per-direction permeability)

If the user enters L_x, L_y, L_z via **LX/LY/LZ** and activates **LTOSIGMA** with coefficients f_x, f_y, f_z (default 4, 4, 4):

$$T_{mf} = \text{CDARCY } V \left(\frac{f_x K_x}{L_x^2} + \frac{f_y K_y}{L_y^2} + \frac{f_z K_z}{L_z^2} \right) \quad (\text{TD 2.56 ; RM 3.116})$$

With the **XONLY** option (default), only the K_x term applies; with **ALL**, all three directions contribute.

If *LTOSIGMA* is present, any explicit *SIGMA(V)* / *SIGMAGD(V)* are ignored.

3 Gravity imbibition / drainage

Gravity head between a matrix and fracture cell (**GRAVDR**, standard)

Let X_f, X_m denote the fractional heights of the water front in the fracture and matrix cells respectively. Integrating the rock-Pc plus gravity head over the cell volume yields the pseudo capillary pressure:

$$\Delta p_{\text{grav}} = \rho_w g \Delta Z_{\text{mat}} (X_f - X_m) \quad (\text{TD 2.57})$$

The total oil flow from fracture to matrix in a gas-oil system becomes

$$q_g = T_{mf} \text{ GMOB} \left[(P_{o,f} - P_{o,m}) + (P_{c,g,f} - P_{c,g,m}) + \frac{1}{2} \Delta Z_{\text{mat}} (X_f - X_m) (\rho_g - \rho_o) g \right] \quad (\text{TD 2.58})$$

$$q_o = T_{mf} \text{ OMOB} \left[(P_{o,f} - P_{o,m}) - \frac{1}{2} \Delta Z_{\text{mat}} (X_f - X_m) (\rho_g - \rho_o) g \right] \quad (\text{TD 2.59})$$

The gravity drainage head is split between phases; the factor $\frac{1}{2}$ comes from the symmetric vertical-equilibrium argument.

Eclipse 300: smooth interpolation of **SIGMA** and **SIGMAGD**

When **SIGMAGD** is supplied, Eclipse 300 uses a per-phase effective sigma:

$$\sigma_p^{\text{eff}} = \omega_p \sigma + (1 - \omega_p) \sigma_{\text{gd}} \quad (\text{TD 2.60})$$

$$\omega_p = \frac{\Delta \theta_{\text{gd}}}{\Delta \theta_{\text{gd}} + \Delta \Phi_{ij}} \quad (\text{TD 2.61})$$

where

$$\Delta \Phi_{ij} = (P_i - P_j) + (P_{c,i} - P_{c,j}) - \rho_p g \Delta D \quad (\text{TD 2.62})$$

The gravity-drainage potential $\Delta\theta_{\text{gd}}$ is phase-specific:

$$\Delta\theta_o = -\frac{1}{2}(\Delta\theta_g^m + \Delta\theta_w^m) \quad (\text{TD 2.63})$$

$$\Delta\theta_g = \frac{1}{2} \Delta\theta_g^m \quad (\text{TD 2.64})$$

$$\Delta\theta_w = \frac{1}{2} \Delta\theta_w^m \quad (\text{TD 2.65})$$

with phase-vs-oil gravity-head differences

$$\Delta\theta_g^m = g (Y_{f,g} - Y_{m,g}) (\rho_g - \rho_o) \quad (\text{TD 2.66})$$

$$\Delta\theta_w^m = g (Y_{f,w} - Y_{m,w}) (\rho_w - \rho_o) \quad (\text{TD 2.67})$$

$Y_{f,g}, Y_{m,g}$ are the fractions of the cell containing mobile gas (water); estimated by a vertical-equilibrium model.

Limit cases: gravity-dominant $\Rightarrow \omega_p \rightarrow 0 \Rightarrow \sigma_p^{\text{eff}} \rightarrow \sigma_{gd}$. *Capillary-dominant* $\Rightarrow \omega_p \rightarrow 1 \Rightarrow \sigma_p^{\text{eff}} \rightarrow \sigma$.

Alternative model (GRAVDRM, Quandalle–Sabathier 1989)

Matrix-fracture flow decomposed into three components:

$$q = q_h + q_{\text{up}} + q_{\text{down}} \quad (\text{TD 2.68})$$

$$q_h = T_\sigma \lambda_h^\uparrow (P_f + P_{c,f} - P_m - P_{c,m}) \quad (\text{TD 2.69})$$

$$q_{\text{up}} = T_{\sigma_{\text{gd}}} \lambda_{\text{up}}^\uparrow [(P_f + P_{c,f} - P_m - P_{c,m}) + (\rho_p - \rho_p^{\text{mat}}) g \Delta Z_{\text{mat}}/2] \quad (\text{TD 2.70})$$

$$q_{\text{down}} = T_{\sigma_{\text{gd}}} \lambda_{\text{down}}^\uparrow [(P_f + P_{c,f} - P_m - P_{c,m}) - (\rho_p - \rho_p^{\text{mat}}) g \Delta Z_{\text{mat}}/2] \quad (\text{TD 2.71})$$

where the saturation-weighted matrix density is

$$\rho_p^{\text{mat}} = \rho_o S_o + \rho_g S_g + \rho_w S_w.$$

T_σ is the matrix-fracture transmissibility built from SIGMA; $T_{\sigma_{\text{gd}}}$ from SIGMAGD.

Re-infiltration toggle (GRAVDRM item 1, YES/NO) controls whether oil may flow back into the matrix. With YES, final recovery depends on transmissibility (non-physical); NO yields a transmissibility-independent equilibrium.

4 Viscous displacement (VISCD, Gilman–Kazemi 1988)

Consider a matrix block at pressure P_m surrounded by a fracture with pressure varying from P_1 (upstream) to P_2 (downstream). Three regimes:

(a) $P_m > P_1, P_2$ — **pure matrix→fracture flow:**

$$q = \frac{T}{2} \lambda_m (P_m - P_1) + \frac{T}{2} \lambda_m (P_m - P_2) = T \lambda_m (P_m - \bar{P}_f) \quad (\text{TD 2.72–2.73})$$

where $\bar{P}_f = (P_1 + P_2)/2$ is the average fracture pressure.

(b) $P_m < P_1, P_2$ — **pure fracture→matrix flow:**

$$q = T \lambda_f (\bar{P}_f - P_m) \quad (\text{TD 2.74})$$

(c) **Mixed:** $P_1 > P_m > P_2$ — **flow in at one end, out at the other:**

$$q = \frac{T}{2} \lambda_f (P_1 - P_m) - \frac{T}{2} \lambda_m (P_m - P_2) \quad (\text{TD 2.75})$$

After algebraic rearrangement using $P_{1,2} = \bar{P}_f \pm (P_1 - P_2)/2$:

$$q = \frac{T}{2} \lambda_f \left[\bar{P}_f + \frac{P_1 - P_2}{2} - P_m \right] - \frac{T}{2} \lambda_m \left[P_m - \bar{P}_f + \frac{P_1 - P_2}{2} \right] \quad (\text{TD 2.76})$$

Define the fracture potential gradient G via $P_1 - P_2 = G L$ (where L is the matrix-block dimension in the relevant direction from LX/LY/LZ):

$$\boxed{q = \frac{T}{2} \lambda_f (\bar{P}_f - P_m) - \frac{T}{2} \lambda_m (P_m - \bar{P}_f) + \frac{L}{4} T (\lambda_f - \lambda_m) G} \quad (\text{TD 2.77})$$

The last term is the explicit viscous-displacement contribution — proportional to the matrix block dimension L and the fracture pressure gradient G .

Estimating G from neighbour cells. Let ϕ_{p-} and ϕ_{p+} be the absolute values of the potential differences to the upstream and downstream adjacent cells, and D_l, D_r the centre-to-centre distances:

$$G = \frac{\phi_{p-}}{2 D_l} + \frac{\phi_{p+}}{2 D_r} \quad (\text{TD 2.78})$$

Computed from previous-timestep potentials.

5 DPKRMOD: matrix oil-Kr modification

For $m_w > 0$, the matrix oil-in-water relative permeability is bent toward the midpoint:

$$k_{r,o}(S) = k_{r,T}(S) + g(S) [k_M - k_{r,T}(S)] \quad (\text{TD 2.79 ; RM 3.59})$$

where

- $k_{r,T}(S)$ is the looked-up rock k_r ;
- $k_M = k_{r,T}(S^*)$ at the midpoint $S^* = (S_{owcr} + (1 - S_{wco}))/2$;
- $g(S) = 4m_w S^2 - 4m_w S + 1$.

End-points and the midpoint value are preserved by construction.

For $m_w < 0$, the implicit (X-Y swapped) version is used: given S , solve

$$S + g(S') (S' - S) = S^* \quad (\text{TD 2.80 ; RM 3.60})$$

with $g(S') = 4m_w S'^2 - 4m_w S' + 1$, then take $k_r = k_{r,T}(S')$.

Fracture-matrix k_r scaling (DPKRMOD item 3 = YES)

The fracture k_r used *only* for fracture-to-matrix flow is scaled so the maximum equals the matrix k_r at the displaced-phase residual:

$$k_{r,w}^{F \rightarrow M} = k_{r,w}^{\text{tab,frac}}(S_w) \cdot \frac{k_{r,w}^{\text{mat,max}}}{k_{r,w}^{\text{frac,max}}} \quad (\text{TD 2.81 ; RM 3.61})$$

with $k_{r,w}^{\text{mat,max}} = k_{r,w}^{\text{mat}}(S_w = 1 - S_{owcr})$. Analogous expressions apply to gas and oil.

6 Linear system structure (Schur reduction)

The Jacobian system for DP / DK in residual form is

$$\begin{pmatrix} A & C \\ D & B \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{r}_m \\ \mathbf{r}_f \end{pmatrix} \quad (2)$$

where $\mathbf{x}$ collects matrix unknowns, $\mathbf{y}$ collects fracture unknowns; A, B are intra-cell Jacobians and C, D are diagonal coupling matrices (each matrix cell connects to exactly one fracture cell).

DUALPORO (A diagonal): exact Schur reduction

Eliminate $\mathbf{x}$ algebraically:

$$(B - D A^{-1} C) \mathbf{y} = \mathbf{r}_f - D A^{-1} \mathbf{r}_m \quad (\text{TD 2.82})$$

$$\mathbf{x} = A^{-1}(\mathbf{r}_m - C \mathbf{y}) \quad (\text{TD 2.83})$$

Because A is diagonal, $D A^{-1} C$ is also diagonal and the band structure of B is preserved. Effective problem size: same as a single-porosity simulation of the fracture grid.

DUALPERM (A banded): simultaneous solve via nested factorisation

Each cell-pair Jacobian block is

$$J_{\text{pair}} = \begin{pmatrix} A_m & D \\ C & B_f \end{pmatrix} \quad (\text{TD 2.84})$$

(A_m, B_f are 3×3 in three-phase). The full Jacobian along x, y, z has 7 bands:

$$J = D + L_1 + U_1 + L_2 + U_2 + L_3 + U_3 \quad (\text{TD 2.85})$$

Nested factorisation builds an approximation $\tilde{J}$ through four levels:

$$\begin{aligned} T &= D + L_3 S^{-1} U_3 \\ S &= T + L_2 R^{-1} U_2 \\ R &= S + L_1 Q^{-1} U_1 \\ Q &= \Gamma + L \Gamma^{-1} U \end{aligned} \quad (\text{TD 2.86})$$

where Γ is diagonal and L, U inherit the band structure of J . The fourth nesting level (innermost) accounts for the matrix-vs-fracture coupling. OPTIONS item 60 > 0 reverts to the pre-97A solver.

7 Initialisation: fractional volumes for gravity drainage

Initial fractional volume of water below the water contact

$$X_{w,i} = \frac{S_{w,i} - S_{w,co}}{1 - S_{cohy} - S_{w,co}} \quad (\text{TD 2.87})$$

with

$$S_{cohy} = \begin{cases} S_{ocow} & (\text{oil/water}) \\ S_{ocow} + S_{gco} & (\text{oil/water/gas}) \\ S_{gco} & (\text{gas/water}) \end{cases}$$

and saturation conditions

$$\begin{aligned} X_w = 1, & \quad S_w = S_{w,\max} = S_{w,i}(1 - S_{cohy}) + (1 - S_{w,i})(1 - S_{cohy}) \\ X_w = 0, & \quad S_w = S_{w,\min} = S_{w,i} S_{w,cr} + (1 - S_{w,i}) S_{w,cr} \end{aligned}$$

For S_w in $(S_{w,\min}, S_{w,\max})$, X_w varies linearly between 0 and 1.

Initial fractional volume of gas above the gas contact

$$X_{g,i} = \frac{S_{g,i} - S_{g,co}}{1 - S_{l,co} - S_{g,co}} \quad (\text{TD 2.88})$$

with $S_{l,co}$ defined analogously to S_{cohy} for the connate liquid combination.

8 Integrated capillary pressure (INTPC)

For a piecewise-linear input table ($S_g \mapsto P_c$), the average gas saturation in a matrix block of height h up to the height where $P_c = P$ is

$$\bar{S}_g = \frac{1}{P} \int_0^P S_g^{-1}(P_c) dP_c \quad (\text{TD 2.89})$$

where S_g^{-1} is the inverse capillary-pressure function.

The gravity-drainage head as a function of the mobile oil saturation, with the fracture full of gas (standard GRAVDR model):

$$P_{cgr} = \frac{S_o - S_{o,r}}{S_{op}} \Delta \rho g h \quad (\text{TD 2.90})$$

with $S_{op} = 1 - S_{wco} - S_{ogcr}$ and $\Delta \rho = \rho_o - \rho_g$.

The constructed (*integrated*) capillary pressure satisfies $P'_c(\bar{S}_g) = P_{cgr}$, so the matrix equilibrium saturation matches the continuous-matrix-medium final recovery:

$$P'_c(\bar{S}_g) = \frac{S_o - S_{o,r}}{S_{op}} P \quad (\text{TD 2.91})$$

where P is the true capillary pressure at the top of the block when the equilibrium saturation is $\bar{S}_g$. Implemented as a tabulated function $\bar{S}_g \leftrightarrow P$ derived from (TD 2.89).

9 Discretized matrix (NMATRIX / NMATOPTS)

The matrix block is partitioned into N nested sub-cells (innermost $n = 1$ near the fracture). Let V_n be the volume of the n -th sub-cell. Total matrix volume:

$$V_T = \sum_{n=1}^N V_n \quad (3)$$

Outer sub-cell volume V_1 is a fraction of the fracture pore volume (NMATOPTS item 2, default 0.1). The remaining sub-cells grow geometrically with factor

$$f = \left(\frac{X}{X_1} \right)^{\frac{1}{N-1}} \quad (4)$$

where X is the centre-to-fracture distance for the whole matrix and X_1 is the outermost sub-cell width. The distance to the inner face of sub-cell n is

$$X_n = f^{n-1} X_1 \quad (5)$$

Geometry-dependent volumes.

- **LINEAR** ($D = 1$): sub-cells are slabs; volume $\propto$ distance.
- **RADIAL** ($D = 2$): sub-cells are cylindrical shells; volume from shell formulae.
- **SPHERICAL** ($D = 3$): sub-cells are spherical shells.

For any geometry, the centre of sub-cell n (for $n > 1$) is

$$X_n^c = \frac{1}{2}(X_{n-1} + X_n) \quad (6)$$

and the interface area between sub-cells n and $n + 1$ scales with the dimensionality:

$$A_n = Z (X - X_n)^{D-1} \quad (7)$$

(Z is a geometry constant: outer-face area for LINEAR; $2\pi \cdot h$ for RADIAL; 4π for SPHERICAL). The original matrix-fracture transmissibility scales to a sub-cell-to-sub-cell transmissibility as

$$T_n = T \cdot \frac{A_n}{A_T} \cdot \frac{2X}{X_n - X_{n-1}} \quad (8)$$

For the outer sub-cell-to-fracture connection, the scale factor is $2X/X_1$. The dimensionality constant Z cancels in any ratio of A_n .

10 Rock-volume sharing in multi-porosity cells

In a multi-porosity cell, each porosity receives a pore volume from input data and a share of the rock volume. Default sharing (TD p.122):

$$V_p^{\text{rock}} = V_p^{\text{pore}} \quad (\text{if total pore volume} \leq \text{available rock volume}) \quad (9)$$

Any rock-volume surplus goes to the matrix; deficit is shared proportionally to pore volume. The keyword **ROCKSPLV** overrides this rule per cell. With **ACTNUM** = 2 (rock-only) or 3 (pore-only), special rules apply — see RM **ACTNUM**.

11 Block-to-block (BTOBALFA) connection

A non-co-located matrix–fracture transmissibility is added between the lower matrix cell and the upper fracture cell. The transmissibility is computed as a standard spatial Darcy connection between cells i and j :

$$T_{ij}^{b2b} = \alpha \text{CDARCY} \frac{2 A_{ij}}{\frac{\Delta D_i}{K_i} + \frac{\Delta D_j}{K_j}} \quad (10)$$

multiplied by the contact-area factor α (BTOBALFA for grid-wide; BTOBALFV for per-cell). Setting α activates the connection; otherwise it does not exist.

Not compatible with multi-porosity (NMATRIX > 1 or TRPLPORO).

12 Summary: full system of equations a DP/DK run solves

For each phase $p \in \{o, g, w\}$ and each cell pair (m, f) :

$$\frac{\partial(\phi S_p)_m}{\partial t} + \nabla \cdot (\rho_p \mathbf{v}_{p,m}) - q_{p,mf} = 0 \quad (\text{matrix balance}) \quad (11)$$

$$\frac{\partial(\phi S_p)_f}{\partial t} + \nabla \cdot (\rho_p \mathbf{v}_{p,f}) + q_{p,mf} - q_{\text{well}} = 0 \quad (\text{fracture balance}) \quad (12)$$

with

- $\mathbf{v}_{p,m} = 0$ in DUALPORO (no matrix-matrix flow); $\mathbf{v}_{p,m}$ from Darcy in DUALPERM.
- $q_{p,mf}$ from (1) (basic) or (TD 2.69)–(TD 2.71) (GRAVDRM) or (TD 2.77) (VISCD).
- Phase potential $\Phi_p = P_o + P_{c,p,o} - \rho_p g D$ with $P_{c,p,o}$ optionally modified by INTPC (TD 2.91).
- Relative permeabilities $k_{r,p}$ optionally modified by DPKRMOD (TD 2.79 ; RM 3.59)–(TD 2.81 ; RM 3.61).
- Sigma σ optionally interpolated between SIGMA and SIGMAGD per (TD 2.60).

plus saturation closure $\sum_p S_p = 1$ and equation-of-state / PVT closure for compositional / black-oil runs.

The Jacobian of this nonlinear system is solved per Newton iteration via (TD 2.82)–(TD 2.83) in DUALPORO (Schur reduction) or via (TD 2.84)–(TD 2.86) in DUALPERM (4-level nested factorisation).

13 Equation → keyword cross-reference

Equation	Concept	Controlling keyword(s)
(TD 2.54)	Matrix-fracture transmissibility	SIGMA, SIGMAV, MULTSIG(V), PERMMF, MU
(TD 2.55 ; RM 3.172)	Kazemi σ	implicit in SIGMA
(TD 2.56 ; RM 3.116)	σ from block dimensions	LTOSIGMA, LX, LY, LZ
(TD 2.57)–(TD 2.59)	Standard gravity drainage	GRAVDR, DZMTRX(V)
(TD 2.60)–(TD 2.67)	E300 σ interp.	SIGMA+SIGMAGD on E300
(TD 2.68)–(TD 2.71)	Quandalle–Sabathier	GRAVDRM, SIGMA, SIGMAGD, DZMTRX
(TD 2.72–2.73)–(TD 2.78)	Viscous displacement	VISCD, LX, LY, LZ, LTOSIGMA
(TD 2.79 ; RM 3.59)–(TD 2.81 ; RM 3.61)	k_r modification	DPKRMOD, KRNUMF, IMBNUMF
(TD 2.82)–(TD 2.83)	DP Schur reduction	automatic under DUALPORO
(TD 2.84)–(TD 2.86)	DK nested factorisation	automatic under DUALPERM; OPTIONS(60)
(TD 2.87)–(TD 2.88)	Initialisation	EQUIL, OPTIONS(11) for steady-state init
(TD 2.89)–(TD 2.91)	Integrated P_c	INTPC
(3)–(8)	Discretized matrix	NMATRIX, NMATOPTS, GRAVDRB (vertical)
(9)	Multi-porosity rock vol.	ROCKSPLV (override)
(10)	Block-to-block	BTOBALFA, BTOBALFV

14 Provenance and deviation notes

- (TD 2.55 ; RM 3.172) Coefficient 4 comes from Kazemi et al. (1976), SPE-5719-PA. Warren & Root (1963) used $\pi^2 \approx 9.87$ for a 1D slab (late-time analytical solution). Lim & Aziz (1995) argue the analytical value should be $\pi^2(n_x/L_x^2 + n_y/L_y^2 + n_z/L_z^2)$ where n_k counts fracture sets per direction; for fully-bounded blocks this gives a coefficient $\pi^2 \approx 9.87$ vs Kazemi’s 4. Eclipse uses Kazemi’s 4.
- (TD 2.69)–(TD 2.71) From Quandalle & Sabathier (1989), SPE Reservoir Engineering 4(4):475–480. GRAVDRM re-infiltration = NO mimics Quandalle–Sabathier’s preferred deployment.
- (TD 2.72–2.73)–(TD 2.77) Gilman & Kazemi (1988), JPT 40(1):60–70, SPE-16010-PA.
- (3)–(8) Discretized matrix is Eclipse’s implementation inspired by the MINC framework of Pruess & Narasimhan (1985), SPE-10509-PA, but using the Kazemi σ -based transmissibility (not MINC’s integral finite difference). Geometrically similar; mathematically distinct in the coupling derivation.
- (TD 2.89)–(TD 2.91) The integration step (TD 2.89) produces a table that is pseudoised differently between GRAVDR and GRAVDRM (2007.1+); see RM INTPC for version notes.

Default values to be aware of.

- SIGMA / SIGMAV default 0 $\Rightarrow$ no matrix-fracture coupling unless LTOSIGMA.
- SIGMAGD / SIGMAGDV default $\rightarrow$ fall back to SIGMA values.
- DZMTRX default 0 $\Rightarrow$ no gravity drainage even when GRAVDR is active.
- Fracture permeability is multiplied by fracture porosity unless NODPPM is set: $K_f^{\text{eff}} = K_f^{\text{input}} \cdot \phi_f$ (without NODPPM); $K_f^{\text{eff}} = K_f^{\text{input}}$ (with NODPPM).

A Equation index (by TD equation number)

TD Eq.	Label in this document
2.54	(TD 2.54)
2.55	(TD 2.55 ; RM 3.172)
2.56	(TD 2.56 ; RM 3.116)
2.57	(TD 2.57)
2.58	(TD 2.58)
2.59	(TD 2.59)
2.60	(TD 2.60)
2.61	(TD 2.61)
2.62	(TD 2.62)
2.63	(TD 2.63)
2.64	(TD 2.64)
2.65	(TD 2.65)
2.66	(TD 2.66)
2.67	(TD 2.67)
2.68	(TD 2.68)
2.69	(TD 2.69)
2.70	(TD 2.70)
2.71	(TD 2.71)
2.72–2.73	(TD 2.72–2.73)
2.74	(TD 2.74)
2.75	(TD 2.75)
2.76	(TD 2.76)
2.77	(TD 2.77)
2.78	(TD 2.78)
2.79	(TD 2.79 ; RM 3.59)
2.80	(TD 2.80 ; RM 3.60)
2.81	(TD 2.81 ; RM 3.61)
2.82	(TD 2.82)
2.83	(TD 2.83)
2.84	(TD 2.84)
2.85	(TD 2.85)
2.86	(TD 2.86)
2.87	(TD 2.87)
2.88	(TD 2.88)
2.89	(TD 2.89)
2.90	(TD 2.90)
2.91	(TD 2.91)

End of mathematical model.